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RESEARCH ARTICLE

Maximum Flow Model With Multiple Origin and Destination and Its Application in Designing Urban Drainage Systems

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ABSTRACT In this paper, a multi-origin and multi-destination maximum flow model is proposed by integrating the objective programming method with graph theory. Through Model 1, low-cost design of drainage pipeline networks can be achieved. For instance, in the experiments, certain pipelines such as Pipeline 9, Pipeline 20, and Pipeline 27 can be excluded during the initial construction phase while still maintaining effective drainage performance and reducing costs. Model 2 enables the monitoring of drainage capacity at various nodes within the existing drainage system. For example, under a specific rainfall intensity, it was observed that certain nodes, such as Nodes 9∼16, Node 45, and Node 48, exhibit poor drainage capacity, necessitating precautionary measures. Model 3 focuses on the expansion planning of the drainage system, ensuring its normal operation under certain rainfall intensities. The results based on a specific area in Changchun City demonstrate that the proposed models not only possess strong practical application value in the design of urban drainage systems but also have broader applicability in other fields such as traffic flow and logistics supply chain planning. This study opens up new avenues for model construction and enriches the practical application effects of mixed-integer programming models.

INDEX TERMS Drainage system planning, maximum flow, objective programming, waterlogging.

I. INTRODUCTION

In recent years, the frequent occurrence of extreme weather events, exacerbated by factors such as global warming, has profoundly impacted the economy, society, life safety, and environmental protection [1], [2], [3]. With the increase in extreme precipitation events, urban waterlogging has become increasingly severe [4], [5], [6]. Urban waterlogging refers to the phenomenon where surface runoff accumulates in large quantities due to insufficient urban drainage capacity during persistent heavy rainfall [7]. In recent years, extreme precipitation disasters have occurred frequently worldwide [8], causing significant local losses [9], and the phenomenon of ''urban sea-viewing'' has become common [10]. The main causes of waterlogging can be attributed to two points: one is the increase in extreme precipitation due to global warming, and the other is the lack of optimization in drainage network design during rapid urbanization.

Faced with the increasingly severe problem of urban waterlogging, some scholars have predicted areas that may experience extreme precipitation in the future [11], [12], [13]. However, the current research still has significant deviations in prediction accuracy, making it unable to provide sufficient basis for future urban flood control measures. Due to the complexity and uncertainty of future extreme precipitation scenarios faced by cities, the renovation of existing urban drainage networks and the construction planning of new urban areas have received increasing attention [14], [15]. Tests on existing urban drainage networks have shown that issues such as aging and structural damage are the main factors limiting their drainage capacity [16]. Drainage networks designed for traditional rainfall scenarios can

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no longer meet the drainage needs of current scenarios, which is also one of the factors leading to frequent urban waterlogging [17]. Therefore, the construction and design of drainage networks in the early stages of urban planning and the reasonable renovation of existing drainage networks have become crucial [18], and the planning and renovation of urban drainage systems have become key issues in adapting to this transformation [19].

As an important component of urban underground infrastructure, drainage networks are usually planned and constructed in the early stages of urban development, determining the system's drainage capacity at that time. Although the drainage capacity can be increased later by replacing pipes with larger diameters, this method is economically inefficient and can cause significant interference to municipal infrastructure and residents' quality of life. Therefore, early optimization of drainage system design is essential to effectively meet future needs. When planning new drainage networks, the following issues must be evaluated: (1) whether the system capacity is sufficient to meet current needs; (2) whether it can effectively mitigate urban waterlogging during extreme rainfall; (3) whether future renovations are economically efficient and feasible.

Scholars have established and refined urban drainage systems based on multi-objective programming theory, integrating construction costs and drainage effectiveness [20], [21]. Some scholars also employ integer nonlinear programming models to plan drainage systems while taking construction costs into consideration [22], [23]. This study develops a systematic pipeline flow detection model and capacity optimization model using a Mixed Integer Programming (MIP) model, based on pipeline flow and capacity, combined with actual pipeline data from a university. The study primarily uses the latitude and longitude coordinates of drainage points to determine specific locations; the elevation of drainage points to determine drainage direction; runoff coefficients and catchment areas to calculate the water collection volume at drainage points; and circular pipe segments with diameters ranging from 400 to 1000 mm to determine the drainage volume per unit time. The established Model 1 in this study constructs an urban drainage system while considering both extreme precipitation response and construction costs. Model 2, without altering the existing drainage system, monitors the drainage capacity of each node in real-time and minimizes overall losses by regulating drainage nodes to respond to extreme precipitation. Model 3 provides specific measures for expanding the existing drainage system, considering both extreme precipitation response and expansion costs. All three models simulate the drainage capacity of pipelines under storm conditions, with the final model effects produced under the premise of preventing extreme precipitation. Compared to other drainage methods, the MIP model used in this study offers greater flexibility in parameter adjustment, allowing for the setting of corresponding parameters for different pipeline types or drainage systems. Additionally, the importance of responding to different storm scenarios or construction costs can also be achieved by adjusting parameters. Furthermore, this experiment uses Matlab2023b with the Gurobi solver for efficient solution of large-scale drainage system networks.

II. CONSTRUCTION PROCESS OF URBAN PIPELINE NETWORK

Water is an important component of urban waterlogging and is subjected to temporal and spatial shifts. A prerequisite for studying the migration process of water is modeling the drainage network [24], [25]. The construction of a drainage network is generally divided into two steps. First, abstract different regions into basic spatial units and second, determine the direction of flow of water between units based on the height differential between them [26] This paper will design a basic urban drainage network based on the above steps. Generally, the simulation of urban drainage networks relies on urban grid images and digital elevation to divide different areas of the city to determine the direction of water flow between areas [27], [28], [29]. To simplify the modeling process and reduce computational complexity, the water flow direction between different regions is defined by height difference, and specific area elevation information is not presented. We abstract the water outlets between different regions into a single point, which helps to showcase the flow trends between different regions. We then bring in specific algorithms for solution and analysis. The process is as shown in Figure 1.

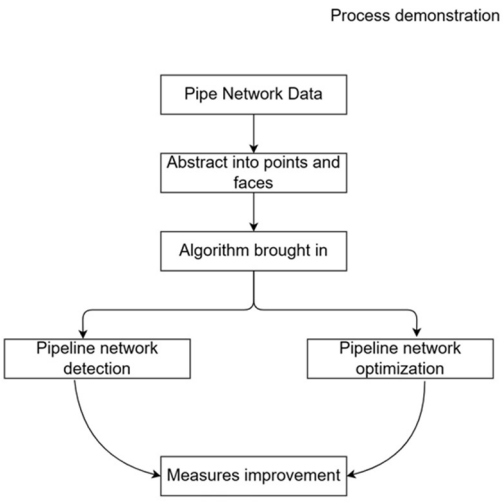


FIGURE 1. Pipeline network optimization process display diagram.

The main approach of this experiment is as follows: Detect the information of each drainage point within the study area, including latitude and longitude information, and altitude information. Utilize the differences in altitude at each point to determine the direction of drainage flow. Simultaneously, based on information such as the runoff coefficient set by the model, calculate the flow information of the catchment area flowing into the drainage points under the intensity of

rainstorm (the runoff coefficient setting refers to Table 1, which is the unified standard for runoff coefficient setting). Table 1 was created based on the *Standard for Design of Building Water Supply and Drainage* (GB50015-2019) of the People's Republic of China. The aforementioned information constitutes the drainage network data in Figure 1. After converting the data into point and surface data, it is input into Model 1, Model 2, and Model 3. An effective drainage system can be established through Model 1, dynamic monitoring of the existing drainage network can be achieved through Model 2, and Model 3 is for the expansion and optimization of the existing drainage network. Among them, the algorithms of Model 1 and Model 2 are detection algorithms for the drainage network, and Model 3 is the optimization algorithm for the drainage network. Through the results of the algorithms, specific improvement measures for the network can be provided.

TABLE 1. Runoff coefficient.

Ground type	Ψ
All kinds of roofing, concrete or asphalt pavement	0.85 ~ 0.95
Gravel pavement with large stone pavement or asphalt surface treatment	0.55 ~ 0.65
Graded gravel pavement	0.40 ~ 0.50
Dry masonry or gravel pavement	0.35 ~ 0.40
Non-paved dirt pavement	0.25 ~ 0.35
Parks or green spaces	0.10 ~ 0.20
COMPREHENSIVE RUNOFF COEFFICIENT	
Regional situation	Ψ
Densely built areas of the city	0.60 ~ 0.85
The city is a densely populated area	0.45 ~ 0.6
Sparse areas of urban architecture	0.20 ~ 0.45

In the official websites of traditional construction of urban drainage, several scholars attribute urban drainage problems to the simple maximum flow problem of a single starting point and a single endpoint in graph theory [30]. Simply put, it involves planning the optimal route between a single drainage inlet and a single outlet. The advantage of this approach is that it can achieve optimal drainage at a single node, however, the actual construction of urban drainage networks requires parallel connections between multiple origin and destination points, and more importantly, considers the overall benefits. In response to the potential large-scale precipitation scenarios in the future, there is urgency to solve the problem of how to achieve the optimal drainage capacity of the urban system under the premise of controllable economic costs. In this paper, a graph theory based maximum flow theory with multiple starting points and endpoints is proposed which is then combined with multi-objective programming with constraints. In contrast to the previous maximum flow problems, which could only solve the problem of a single origin and destination points, our theory can comprehensively analyze multiple origin and destination points. It can not only judge whether the existing drainage network in a city can cope with future large-scale rainfall, but can also optimize and improve the existing urban network. This theory not only effectively solves urban drainage problems in reality, but also enriches the maximum flow theory in graph theory to a certain extent.

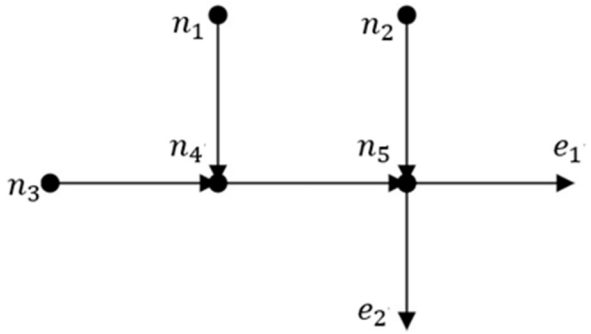


FIGURE 2. Multi origin and multi destination in the maximum flow.

Figure 2 shows the idea of this theory.$\{e_1e_2\} \in E, E$ represents all water outlets. $\{n_1n_2\ n_3n_4n_5\} \in N, N$ represents all water inlet nodes. The traditional maximum flow algorithm can only calculate the maximum flow rate from a single inlet to a single outlet, such as $n_3 \longrightarrow e_1$, The model developed in this paper can calculate the maximum flow rate between system N and system E, forming a merged network, which has a good guiding role for urban pipeline network construction.

III. MODEL INTRODUCTION

In this section, we formulate the problem, and design the model, both ideal and dynamic. We then introduce our optimal expansion model for drainage network.

A. PROBLEM FORMULATION

The purpose of optimizing the design of urban drainage systems is to address the potential urban waterlogging caused by future extreme precipitation. When extreme weather occurs and the precipitation exceeds the load of the drainage system itself, the ground will be subjected to waterlogging.

A loss function can be established to describe the extent of damage to the city. Transforming the drainage system is a better response, but it will also entail huge investment. Therefore, the key to designing drainage networks is the optimal selection of construction sites and specifications, as well as the balance between construction costs and social benefits. This paper adopts the design approach shown in Figure 3 to establish different objective functions and constraints for solving urban drainage network design problems in different contexts. Next, the algorithm developed in this paper is applied. We, then, analyze the problem using operational data, and display the results.

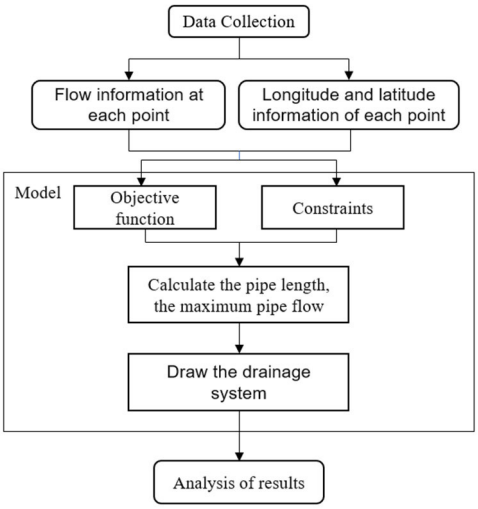


FIGURE 3. The process of urban drainage systems.

We make the following assumptions in our methodology:

- Rainstorm intensity, runoff coefficient and area of the catchment area are assumed to be constant
- Based on available research data [31], [32], [33], The relationship between the diameter of the pipe and the price range of the pipe is assumed to be linear
- The flow velocity in the pipeline is assumed to have multiple waves within a small range, without considering variables
- The shape of the pipe network is uniformly circular
- Unified standard values for slope setting in different areas
- The K-th catchment area corresponds to the K-th node
- The maximum flooding safety limit for a single catchment area is set to 0.9

B. MODEL DESIGN

1) IDEAL PIPE NETWORK DESIGN MODEL

In the planning and construction of the drainage pipe network of new cities, the overall optimal drainage system can be established through the linear programming model. When designing drainage systems in such cities, the main consideration is the quantitative relationship between the unit time drainage volume and the catchment volume at key locations, which can be transformed into an idealized drainage network capacity design problem.

We develop a design model for ideal urban network, based on the unit time catchment of key locations in the city and the flow direction between different regions, combined with relevant knowledge of graph theory such as shortest path theory, maximum flow theory, and optimization.

In the process of optimizing drainage network design, the elevation between drainage nodes can be adjusted for optimization, but the cost would be too high and the cycle would be long. Therefore, controlling the pipeline flow between drainage nodes is often a better solution. We establish the following idealized drainage pipeline design model for this purpose.

a: MODEL 1

The objective function of Model 1 is to minimize the construction cost of the underground drainage system. In this model:

- T : represents the investment repayment period.
- e : denotes the annual depreciation and maintenance rate.
- $C_{1,j}$: indicates the construction cost per unit length of the *j-th* pipe segment, measured in yuan per meter.
- L_j : is the length of the *j-th* pipe segment.

Thus, the objective function is formulated as follows:

$$Min\left\{\left(\frac{1}{T}+e\right)\sum_{j\in J}C_{1,j}L_j\right\}$$

Constraints

1.Flow Capacity Constraint

The first constraint ensures that the total drainage flow through each pipe segment j does not exceed its maximum capacity. Specifically:

$$\sum\nolimits_{i\in N(j)} x_i \le \beta D_j^{\frac{8}{3}}, j \in J \tag{1a}$$

Where:

- x_i : represents the set of drainage paths for the *i-th* route.
- $N(j)$: is the set of all drainage routes that pass through the *j-th* pipe.
- D_j : represents the diameter of the *j-th* pipe segment.
- β : The coefficient for converting pipe diameter to the maximum drainage capacity of the pipe segment.

2.Node Drainage Capacity Constraint

The second constraint ensures that the maximum drainage capacity of each node k is determined by the sum of flows along all drainage routes originating from that node:

$$\sum\nolimits_{i=k} x_i = Q_k, \quad k\in K \tag{1b}$$

where:

- Q_k : is the maximum drainage flow rate at node k per unit time.

3.Catchment Area Constraint

The third constraint defines the maximum drainage volume for each node k based on the area of the catchment, the runoff coefficient, and rainfall intensity:

$$C_k = \psi_k q F_k, \quad k \in K \tag{1c}$$

where:

- C_k : is the maximum catchment volume for node k per unit time.
- ψ_k : is the runoff coefficient for the sub-catchment of node k.
- q : represents the rainfall intensity, measured in $L/(s{\cdot}ha)$.
- F_k : is the area of the sub-catchment of node k, measured in hectares.

4.Extreme Rainfall Constraint

The fourth constraint establishes that, under ideal conditions, the maximum drainage flow rate per unit time at node k should equal the catchment volume under extreme rainfall conditions:

$$Q_k = C_k, \quad k \in K \tag{1d}$$

5.Diameter Cost Conversion Constraint

The fifth constraint converts the diameter of each pipe segment into a unit construction cost:

$$C_{1,j} = \gamma D_j, \quad j{\in}J, x_i \geq 0, D_j{\geq} 0 \tag{1e}$$

where:

- γ : is the diameter-cost conversion coefficient.

The above constraints ensure that the maximum drainage capacity per unit time at each node equals the flow volume under extreme rainfall scenarios, while also optimizing the overall construction cost of the drainage network. Based on the first and second constraints, the optimal pipe diameters are calculated, ultimately determining the most cost-effective construction plan for the drainage network.

2) DYNAMIC PIPE NETWORK DRAINAGE MODEL

For newly built cities, optimization design can be carried out under an ideal model. However, most cities have already established a certain scale of underground drainage systems. Without renovating the existing pipeline network, we need to establish appropriate constraints and plan the drainage route of each node intelligently to maximize the drainage capacity of key nodes.

a: MODEL 2

The objective function of Model 2 is to minimize the losses caused by overflow at critical nodes to the greatest extent possible. In this model:

- α_k : is the loss assessment coefficient for the *k-th* catchment area.
- S_k^- : represents the negative deviation between the catchment inflow and drainage outflow at the *k-th* node.

Thus, the objective function is formulated as follows:

$$Min \sum_{k \in K} \alpha_k S_k^-$$

Constraints

1.Flow Capacity Constraint

The first constraint ensures that the total drainage flow through each pipe segment j does not exceed its maximum capacity. Specifically:

$$\sum\nolimits_{i \in N(j)} x_i \leq \beta D_j^{\frac{8}{3}}, \quad j \in J \tag{2a}$$

2.Node Drainage Capacity Constraint

The second constraint ensures that the maximum drainage capacity of each node k is determined by the sum of flows along all drainage routes originating from that node:

$$\sum\nolimits_{i=k} x_i = Q_k, \quad k{\in}K \tag{2b}$$

3.Catchment Area Constraint

The third constraint defines the maximum drainage volume for each node k based on the area of the catchment, the runoff coefficient, and rainfall intensity:

$$C_k = \psi_k q F_k, \quad k \in K \tag{2c}$$

4.Catchment-Drainage Balance Constraint

The fourth constraint ensures a balance between the catchment inflow and drainage capacity at node k by minimizing the negative deviation S_k^-:

$$C_k - Q_k - S_k^- = 0, \quad k{\in}K \tag{2d}$$

5.Drainage Safety Threshold

The fifth constraint sets a safety threshold for the drainage capacity at each node to ensure a minimum level of functionality:

$$S_k^- \leq 0.9 C_k, \quad k \in K \tag{2e}$$

6.Non-negativity Constraint

The sixth constraint ensures that flow variables and capacities are non-negative:

$$x_i \geq 0, Q_k \geq 0, S_k^- \geq 0 \tag{2f}$$

The purpose of Model 2 is to dynamically adjust the drainage capacities of critical nodes in real-time, based on the varying significance of each node (α_k) and the scale of precipitation events. This model helps ensure that the drainage network can handle varying rainfall intensities with minimized overflow losses at critical points. By setting dynamic thresholds and balancing catchment inflows with drainage outflows, Model 2 enhances the resilience and efficiency of the drainage system, thereby reducing overall system losses under extreme weather conditions.

3) OPTIMAL EXPANSION MODEL FOR DRAINAGE NETWORK

When the rainfall scale reaches a certain level and the intelligent planning of drainage routes is not effective, it is necessary to consider expanding the key drainage network. The following model can achieve the corresponding effect.

a: MODEL 3

The objective function of Model 3 is to minimize the expansion costs and the overflow losses at critical nodes. In this model:

- $C_{2,j}$: represents the expansion cost per unit length of the *j-th* segment, measured in yuan per meter.
- $opt1$: represents the objective function value of Model 1 (cost minimization).
- $opt2$: represents the objective function value of Model 2 (overflow loss minimization).

Thus, the objective function is formulated as follows:

$$Min\left\{\frac{\left(\frac{1}{T}+e_1\right)\sum_{i\in I} C_{2,j}L_j}{2*opt1}+\frac{\sum_{k\in K}\alpha_k S_k^-}{2*opt2}\right\}$$

Constraints

1. Flow Capacity Constraint

The first constraint ensures that the total drainage flow through each pipe segment jjj after expansion does not exceed its adjusted maximum capacity:

$$\sum\nolimits_{i\in N(j)} x_i \le \beta\left(D_j+y_j\right)^{\frac{8}{3}}, \quad j\in J \tag{3a}$$

where:

- D_j : Diameter of the j-th pipe segment before expansion.
- y_i : Diameter increment due to pipeline expansion.

2. Node Drainage Capacity Constraint

The second constraint defines the drainage capacity Q_k for each node k after expansion:

$$\sum\nolimits_{i=k} x_i = Q_k, \quad k\in K \tag{3b}$$

3. Catchment Area Constraint

The third constraint defines the maximum inflow C_k for each node k, based on the catchment area, runoff coefficient, and rainfall intensity:

$$C_k = \psi_k q F_k, \quad k\in K \tag{3c}$$

4. Expansion Cost Conversion Constraint

The fourth constraint calculates the expansion cost for each pipe segment:

$$C_{2,j} = \gamma y_j, \quad j\in J \tag{3d}$$

5. Catchment-Drainage Balance Constraint

The fifth constraint ensures the balance between catchment inflow and drainage outflow for each node k, accounting for overflow:

$$C_k - Q_k - S_k^- = 0, \quad k\in K \tag{3e}$$

6. Overflow Safety Threshold

The sixth constraint sets an overflow safety threshold, ensuring that the negative deviation S_k^- remains within an acceptable range:

$$S_k^- \le 0.9C_k, \quad k\in K \tag{3f}$$

7. Non-Negativity Constraint

The final constraint ensures non-negativity for the flow variables and pipe expansion parameters:

$$x_i \ge 0, \quad Q_k \ge 0, \quad y_j \ge 0, \quad S_k^- \ge 0 \tag{3g}$$

Model 3 integrates pipeline expansion planning with overflow loss minimization at critical nodes. The first two constraints calculate the post-expansion drainage capacity for individual nodes, while the remaining constraints aim to minimize ground overflow. By incorporating multi-objective optimization, the model enables real-time, balanced decision-making to address both cost efficiency and system resilience during varying precipitation scenarios.

IV. CASE STUDY

In this section, we report the results from a number of numerical studies that we conducted using real life data from an urban area. We applied the data to the above models that we developed. These results will demonstrate the use of our models, their usefulness for practical applications and will generate meaningful managerial guidelines.



FIGURE 4. Regional schematic diagram.

A. STUDY AREA

Changchun City is located at 43°05′ ∼ 45°15′N and124°18′ ∼ 127°05′ E, and is situated in the northern temperate zone of the mid latitude of the Northern Hemisphere. The climate of Changchun City belongs to the temperate continental semi humid monsoon climate type. In spring, there are many windy weather conditions, with a maximum wind speed of up to 30 meters per second. In summer, southeast winds prevail, with an average temperature of 21.9 degrees Celsius with frequent short-term heavy rainfall.

In autumn, the weather is sunny and warm that lasts for several days, with a large temperature difference and lower wind speeds compared to spring. In winter, the weather is clear, dry, and the wind is weak, with an average temperature of minus 12 degrees Celsius and there is prolonged large-scale snowfall. The specific research area is a certain campus located in the southeast of Changchun City. The campus contains various terrain areas, including teaching buildings, dormitory buildings, playgrounds, parks, and other different ground types, which can better simulate various complex surface scenarios in real urban construction.

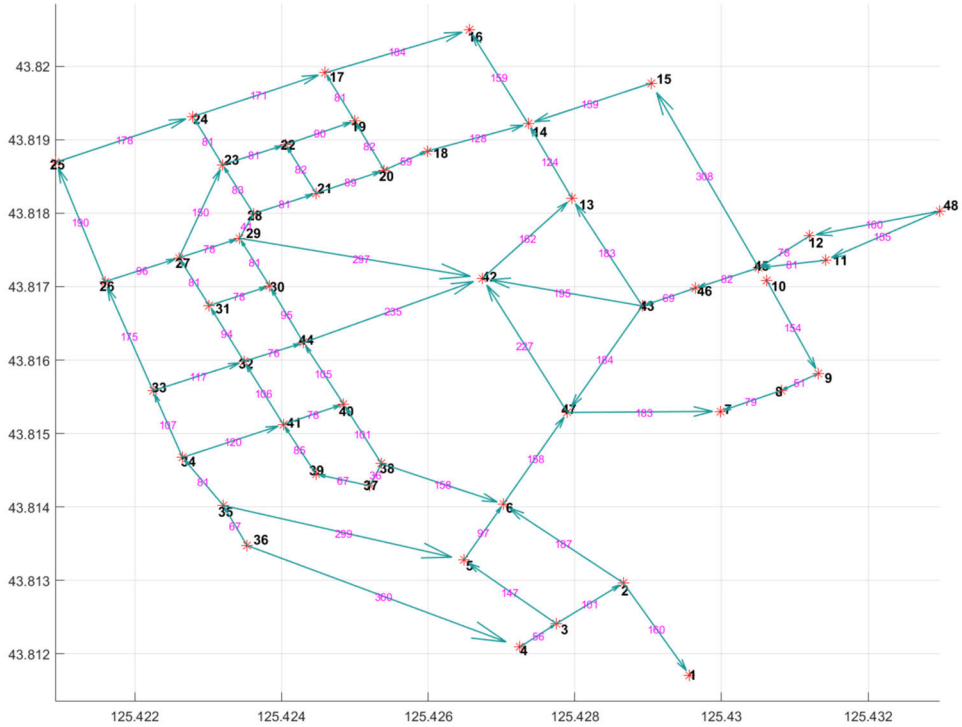


FIGURE 5. Actual drainage network information.

B. SIMULATION OF DRAINAGE NETWORK

We use surveying and mapping instruments to measure the altitude and latitude coordinates of key nodes on campus. Further, we determined the length between nodes to simulate pipeline length, and used the height difference between nodes to determine the direction of water flow. As shown in Figure 5. 48 nodes have been marked as drainage nodes in this area, each corresponding to a catchment area. The horizontal and vertical coordinates of a node represent the longitude and latitude of the location where the node is located, respectively. The total area of these 48 catchment areas is 58 hectares, and the drainage network consists of 70 circular pipes with diameters ranging from 400 to 1000mm. The total length of the drainage network is approximately 9000 meters. Due to the terrain being high in the south and low in the north, which is conducive to rainwater discharge, the pipeline network relies entirely on gravity for operation. The difference between the highest and lowest points is 20.72 meters.

Figure 5 displays the drainage system of the area collected on-site, where the red asterisks mark the location of each drainage node. The blue lines represent the drainage network, and the numbers above the blue lines indicate the length of the pipe sections.

C. DATA PREPARATION

For our simulation, we need a formula for rainstorm intensity. This formulation is different in different regions. The formulation of rainstorm intensity formula that we use in this paper is adopted from Jilin Architectural Design Institute(https://www.renrendoc.com/paper/227730141.html), and is as follows:

$$q = \frac{896(1 + 0.68\lg P)}{t^{0.6}} \tag{4}$$

In formula (4), P represents the design return period (in years), and t represents the duration of rainfall (in minutes).

D. RESULTS

1) IDEAL PIPE NETWORK DESIGN MODEL

Using the first model to establish an idealized pipeline network drainage design, the water collection at each node is brought into the model to calculate the actual flow rate of the idealized pipeline network. Then, based on the actual flow rate, the optimal construction specifications of the drainage pipeline can be calculated.

In Figure 6, the first number in parentheses represents the pipeline number, and the second number represents the pipeline flow rate in the ideal state. The green part indicates that the flow velocity of the pipeline is zero, and the blue part indicates that there is water flowing through the pipeline network. From the figure, it can be seen that pipelines 27, 9, and 20 are located in the middle of the area, with long lengths and high construction costs. However, in the ideal operation of the pipeline network, there is no water flow. Therefore, in actual planning, this part of the pipeline network can be

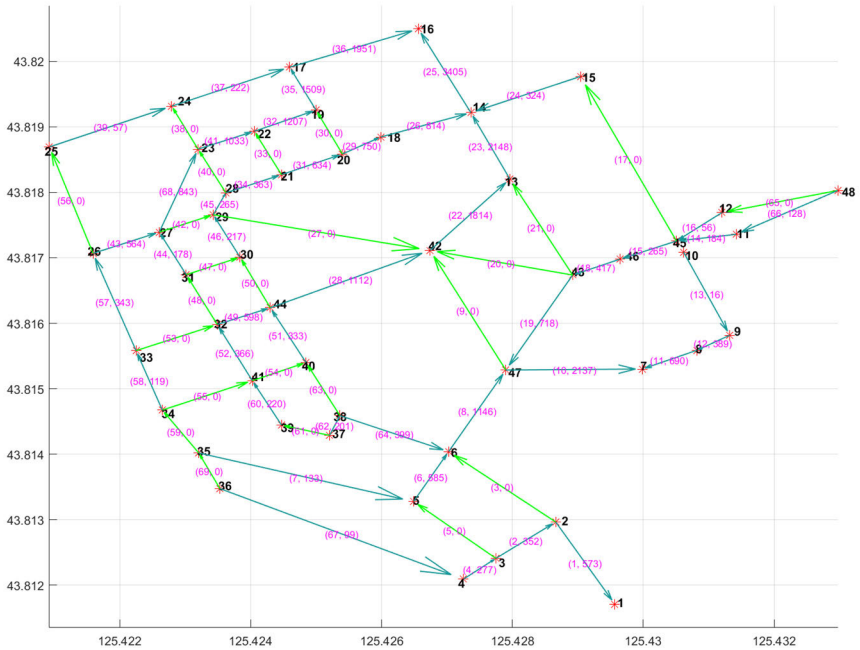


FIGURE 6. Schematic diagram of optimal pipeline network construction.

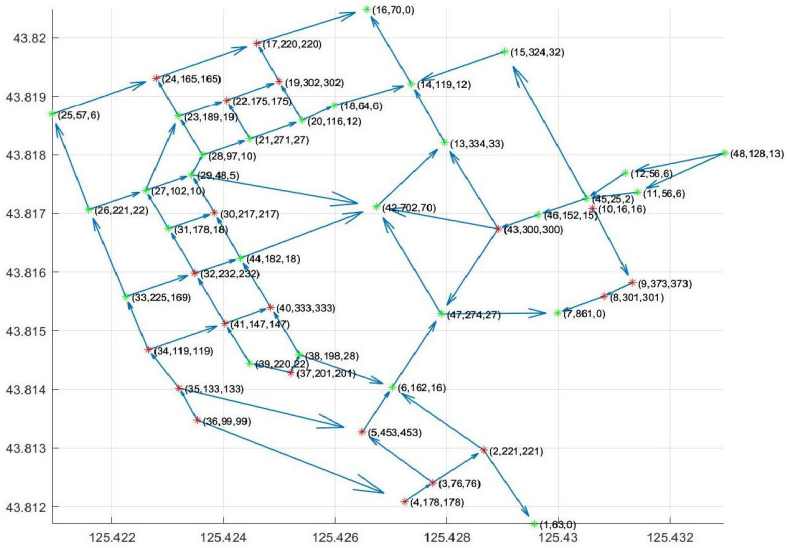


FIGURE 7. Schematic diagram of node drainage.

removed, greatly saving construction costs. In addition, there is also excessive construction in areas with dense pipeline networks, such as pipelines 48, 47, and 42, which can be removed.

2) DYNAMIC PIPE NETWORK DRAINAGE MODEL

For a dynamic pipeline network planning model, the precipitation data with a five-year return period is used for simulation, as shown in Figure 7. There are three numbers in parentheses in the figure. The first number represents the node number, the second number represents the node catchment flow, and the third number represents the node drainage flow. Red represents key nodes, while green represents non key nodes.

From Figure 7, it can be seen that the planning of drainage routes ensures the minimum overflow at some key nodes, while other nodes have a smaller drainage volume while ensuring safety, and the bearing capacity of the pipeline is reasonably allocated. For example, in Figure 7, key nodes such as 43, 9, and 40 have the maximum drainage capacity,

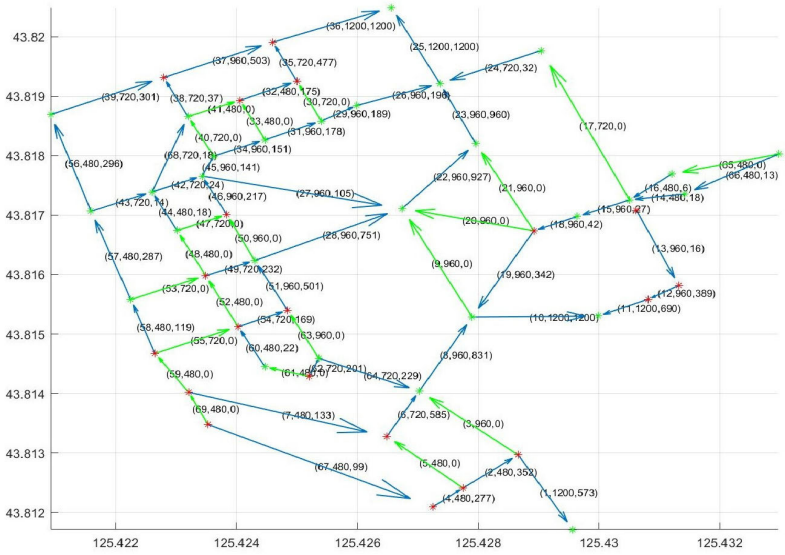


FIGURE 8. Schematic diagram of drainage network flow.

and there is no risk of overflow. Although non-critical nodes, such as 28 and 21, may face overflow risks, the associated losses are relatively minor compared to those at critical nodes. As a result, the overall waterlogging losses of the system will be minimized.

Figure 8 shows the actual flow rate of each pipeline network under the once in five-year precipitation scenario. The first number in parentheses represents the pipeline number, the second number represents the maximum feasible flow velocity of the pipeline, and the third number represents the actual flow velocity of the pipeline. Compared to the idealized pipe network model, as shown in Figure 6, the number of pipes with water flow has increased, such as pipes 56 and 37. This is because under the premise of maintaining the scale of the existing pipeline network, it is impossible to ensure that all nodes do not have overflow problems, and can only maximize the risk of overflow at key nodes. Therefore, more drainage lines will be planned to reasonably allocate the drainage capacity of each node, and the number of pipes through which water flows will also increase.

Figure 9 shows the changes in drainage capacity at different nodes facing rainfall once every five years, once in ten years, and once in twenty years, with the existing pipeline scale unchanged. From the graph, it can be seen that the drainage capacity of some nodes increases, while the drainage capacity of some nodes decreases. For example, the drainage capacity of nodes 32 and 37 decreases significantly with increasing rainfall, while the drainage capacity of nodes 9, 17, and 19 increases with increasing rainfall. This means that in scenarios where rainfall increases, model planning is used to increase the drainage capacity of key nodes while reducing the drainage capacity of non-key nodes to minimize overall losses due to urban waterlogging. The planning of node

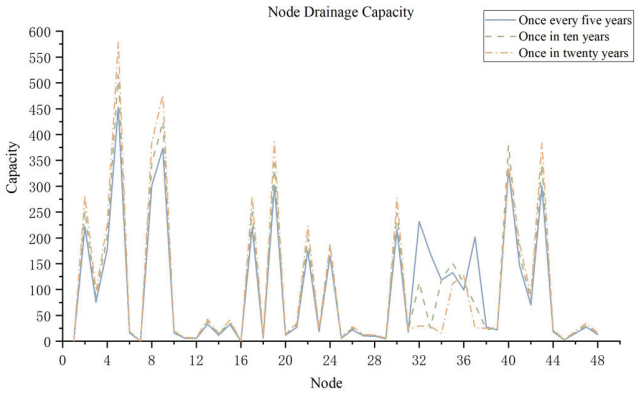


FIGURE 9. Drainage capacity of each node under different rainfall scenarios.

drainage routes is controlled through underground pipeline network valves and pump stations, combined with real-time precipitation data to achieve intelligent control of urban drainage network, which plays an important role in reducing losses due to urban waterlogging.

3) OPTIMAL EXPANSION MODEL FOR DRAINAGE NETWORK

By using precipitation data with a five-year return period and incorporating it into Model 3, we obtain the result shown in Figure 10. The first number in parentheses represents the number of pipelines, the second number represents the maximum carrying capacity of the current pipeline, and the third number represents the scale of the pipeline network that needs to be expanded to cope with the current precipitation conditions. The parts marked in red in the figure represent the pipelines that need to be expanded. The thicker the red pipeline, the larger the size that needs to be expanded. Among

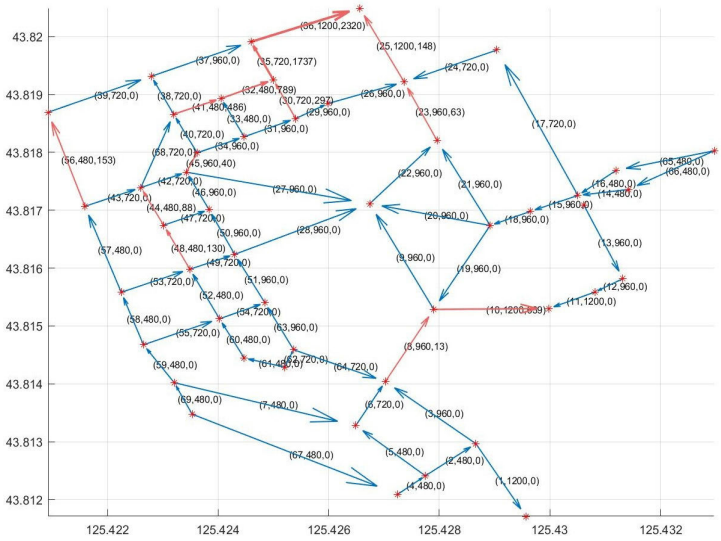


FIGURE 10. Schematic diagram of pipeline network expansion.

them, pipelines 35 and 36 near the water outlet need to be expanded with the largest size, and it also indicates that the pipelines near the water outlet are most prone to congestion problems.

From the result shown in the figure 10, it can be concluded that in order to meet the drainage needs of a certain scale precipitation scenario, only the key drainage network needs to be expanded. This model can help municipal builders analyze the location and expansion scale of important pipeline networks, reduce the scale of urban waterlogging, and save construction costs. This model, therefore, will significantly help urban planners for practical engineering construction.

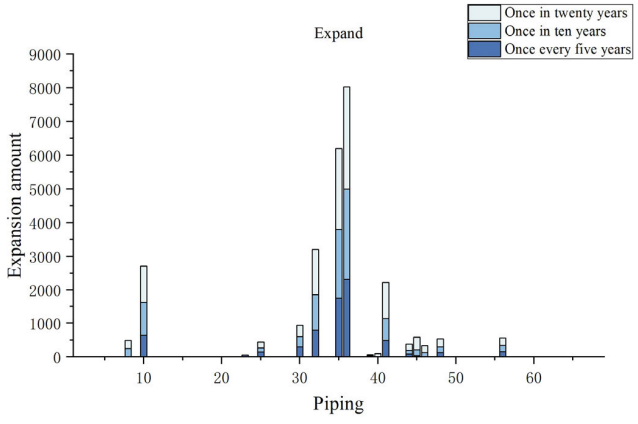


FIGURE 11. Comparison diagram of pipe network expansion.

From Figure 11, it can be seen that with an increase in precipitation, the overall expansion scale of the pipeline network shows an upward trend, but there is no significant increase in the number of pipelines that needs to be expanded. The importance of key drainage network construction has been further clarified. When designing drainage network renovation projects to cope with large rainfall scales, only key network renovations need to be carried out.

V. DISCUSSION AND CONCLUSION

In this study, the actual drainage system was ultimately determined using the latitude and longitude, elevation information of drainage points, and pipeline diameter information, as illustrated in Figure 5. Utilizing Model 1, which prioritizes cost considerations, and incorporating extreme precipitation data, it was calculated that pipelines 9, 20, 27, 42, 47, and 48 could be removed. Using Model 2, the drainage capacity of each node was tested under storm intensities with return periods of 5 years, 10 years, and 20 years, respectively. The specific results are shown in Figure 9, indicating that nodes 1, 9∼16, 25∼29, 45, and 48 have lower drainage capacities and require heightened attention during storm events to implement timely drainage measures. Using Model 3, the existing drainage pipelines were expanded under the same storm intensities, with specific results shown in Figures 10 and 11. Pipelines 10, 32, 35, 36, and 41 were identified as requiring particular focus.

Against the backdrop of increasingly complex extreme climate changes in the future, this study integrates multi-objective programming, graph theory, and drainage network construction practices to establish a maximum flow model with multiple destinations. The model was validated using actual drainage network data from a location in Changchun, demonstrating strong innovative value and practical significance. This study designed an urban drainage network optimization model, an optimal drainage route planning model, and a pipeline network optimization improvement model to address urban waterlogging under varying precipitation scales, showcasing broad applicability. In response to traditional drainage network planning challenges such as unscientific design, insufficient funding, inefficient

TABLE 2. Catchment data.

Node	Runoff Coefficient	Catchment (ha)	Rainstorm intensity (q) P=5, t=10	Confluence flow (L/S)	Rainstorm intensity(q) P=10, t=10	Confluence flow (L/S)	Rainstorm intensity (q) P=10, t=20	Confluence flow (L/S)
1	0.15	1.256	332.038	62.56	378.12	71.24	424.18	79.92
2	0.55	1.21	332.038	220.97	378.12	251.64	424.18	282.29
3	0.65	0.35	332.038	75.54	378.12	86.02	424.18	96.5
4	0.65	0.823	332.038	177.62	378.12	202.28	424.18	226.92
5	0.75	1.818	332.038	452.73	378.12	515.57	424.18	578.37
6	0.2	2.432	332.038	161.5	378.12	183.92	424.18	206.32
7	0.8	3.24	332.038	860.64	378.12	980.09	424.18	1099.47
8	0.8	1.134	332.038	301.22	378.12	343.03	424.18	384.82
9	0.8	1.404	332.038	372.95	378.12	424.7	424.18	476.44
10	0.2	0.238	332.038	15.81	378.12	18	424.18	20.19
11	0.35	0.481	332.038	55.9	378.12	63.66	424.18	71.41
12	0.35	0.484	332.038	56.25	378.12	64.05	424.18	71.86
13	0.4	2.511	332.038	333.5	378.12	379.78	424.18	426.05
14	0.2	1.787	332.038	118.67	378.12	135.14	424.18	151.6
15	0.6	1.628	332.038	324.33	378.12	369.35	424.18	414.34
16	0.25	0.846	332.038	70.23	378.12	79.97	424.18	89.71
17	0.8	0.828	332.038	219.94	378.12	250.47	424.18	280.98
18	0.2	0.957	332.038	63.55	378.12	72.37	424.18	81.19
19	0.8	1.136	332.038	301.76	378.12	343.64	424.18	385.49
20	0.4	0.875	332.038	116.21	378.12	132.34	424.18	148.46
21	0.4	2.043	332.038	271.34	378.12	309	424.18	346.64
22	0.75	0.702	332.038	174.82	378.12	199.08	424.18	223.33
23	0.4	1.424	332.038	189.13	378.12	215.38	424.18	241.61
24	0.7	0.712	332.038	165.49	378.12	188.46	424.18	211.41
25	0.2	0.855	332.038	56.78	378.12	64.66	424.18	72.53
26	0.7	0.95	332.038	220.81	378.12	251.45	424.18	282.08
27	0.4	0.765	332.038	101.6	378.12	115.7	424.18	129.8
28	0.35	0.838	332.038	97.39	378.12	110.9	424.18	124.41
29	0.35	0.414	332.038	48.11	378.12	54.79	424.18	61.46
30	0.8	0.818	332.038	217.29	378.12	247.44	424.18	277.58
31	0.7	0.7655	332.038	177.92	378.12	202.62	424.18	227.3
32	0.7	0.997	332.038	231.73	378.12	263.89	424.18	296.04
33	0.7	0.966	332.038	224.52	378.12	255.68	424.18	286.83
34	0.7	0.51	332.038	118.54	378.12	134.99	424.18	151.43
35	0.7	0.571	332.038	132.72	378.12	151.13	424.18	169.54
36	0.4	0.745	332.038	98.95	378.12	112.68	424.18	126.41
37	0.4	1.516	332.038	201.35	378.12	229.29	424.18	257.22
38	0.4	1.487	332.038	197.5	378.12	224.91	424.18	252.3

TABLE 2. *(Continued.)* **Catchment data.**

39	0.4	1.653	332.038	219.54	378.12	250.01	424.18	280.47
40	0.7	1.431	332.038	332.6	378.12	378.76	424.18	424.9
41	0.7	0.631	332.038	146.66	378.12	167.02	424.18	187.36
42	0.4	5.286	332.038	702.06	378.12	799.5	424.18	896.89
43	0.7	1.292	332.038	300.3	378.12	341.97	424.18	383.63
44	0.4	1.369	332.038	181.82	378.12	207.06	424.18	232.28
45	0.2	0.373	332.038	24.77	378.12	28.21	424.18	31.64
46	0.4	1.147	332.038	152.34	378.12	173.48	424.18	194.61
47	0.4	2.06	332.038	273.6	378.12	311.57	424.18	349.52
48	0.4	0.965	332.038	128.17	378.12	145.95	424.18	163.73

TABLE 3. **Pipe network data.**

Number	Begin	End	Length (m)	Pipe network Diameter (mm)	Expansion (5 years)	Expansion (10 years)	Expansion (20 years)
1	2	1	160	1000	0	0	0
2	3	2	101	400	0	0	0
3	2	6	187	800	0	0	0
4	4	3	56	400	0	0	0
5	3	5	147	400	0	0	0
6	5	6	97	600	0	0	0
7	35	5	299	400	0	0	0
8	6	47	158	800	13	233	242
9	47	42	227	800	0	0	0
10	47	7	183	1000	639	974	1097
11	8	7	79	1000	0	0	0
12	9	8	51	800	0	0	0
13	10	9	154	800	0	0	0
14	11	45	81	400	0	0	0
15	45	46	82	800	0	0	0
16	12	45	78	400	0	0	0
17	45	15	308	600	0	0	0
18	46	43	69	800	0	0	0
19	43	47	184	800	0	0	0
20	43	42	195	800	0	0	0
21	43	13	183	800	0	0	0
22	42	13	162	800	0	0	0

temporary measures, and neglect of human-centric considerations, this study proposes solutions including unified design standards, interdisciplinary integration, and dynamic optimization strategies, providing a theoretical foundation for the construction of intelligent urban drainage networks. Furthermore, the multi-origin, multi-destination maximum flow theory proposed in this study is not only applicable to urban drainage issues but also extends to fields such as information

TABLE 3. *(Continued.)* **Pipe network data.**

23	13	14	124	800	63	0	0
24	15	14	159	600	0	0	0
25	14	16	159	1000	148	125	165
26	18	14	128	800	0	0	0
27	29	42	297	800	0	0	0
28	44	42	235	800	0	0	0
29	20	18	59	800	0	0	0
30	20	19	82	600	297	310	324
31	21	20	89	800	0	0	0
32	22	19	90	400	789	1068	1350
33	21	22	82	400	0	0	0
34	28	21	81	800	0	0	0
35	19	17	81	600	1737	2065	2395
36	17	16	184	1000	2320	2673	3029
37	24	17	171	800	0	0	0
38	23	24	81	600	0	0	0
39	25	24	178	600	0	43	24
40	28	23	83	600	0	0	94
41	23	22	81	400	486	645	1095
42	27	29	78	600	0	0	0
43	26	27	96	600	0	0	0
44	31	27	81	400	88	104	194
45	29	28	41	800	40	173	370
46	30	29	81	800	0	127	211
47	31	30	78	600	0	0	0
48	32	31	94	400	130	167	237
49	32	44	76	600	0	0	0
50	44	30	95	800	0	0	0
51	40	44	105	800	0	0	0
52	41	32	106	400	0	0	0
53	33	32	117	600	0	0	0
54	41	40	78	600	0	0	0
55	34	41	120	600	0	0	0
56	26	25	190	400	153	189	218
57	33	26	175	400	0	0	0
58	34	33	107	400	0	0	0
59	35	34	81	400	0	0	0
60	39	41	85	400	0	0	0
61	37	39	67	400	0	0	0
62	37	38	36	600	0	0	0

TABLE 3. *(Continued.)* **Pipe network data.**

63	38	40	101	800	0	0	0
64	38	6	158	600	0	0	0
65	48	12	160	400	0	0	0
66	48	11	155	400	0	0	0
67	36	4	360	400	0	0	0
68	27	23	150	600	0	0	0
69	36	35	67	400	0	0	0

transmission, logistics scheduling, and power transmission, demonstrating strong interdisciplinary potential.

APPENDIX

See Table 2 and 3.

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